

Spotting Accuracy Issues in Profile-Guided Optimization

A new methodology to spot metadata propagation errors within optimization pipelines

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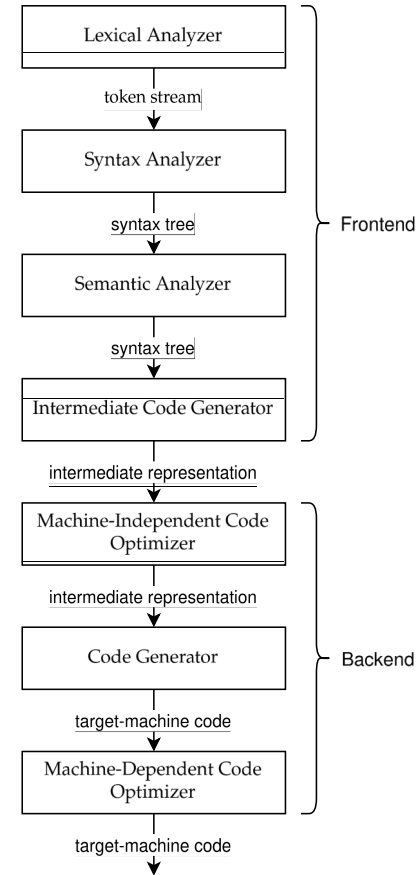
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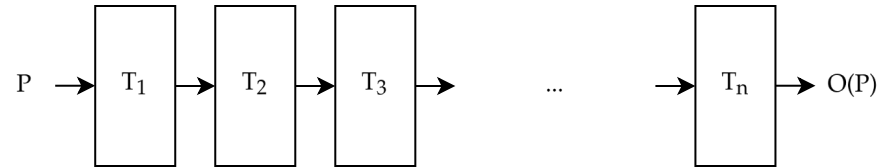
Compilers

- What is a compiler?
 - Translates source code -> target code
 - Preserves program semantic
 - *Optimizes* code during translation
- Compiler Structure (3 Stages)
 - Front-end -> Parses source code -> builds IR
 - Middle-end -> Optimizes IR
 - Back-end -> Generates target code
- Intermediate Representation (IR)
 - Multiple IR types exist
 - Choice depends on compiler design goals



Optimizations

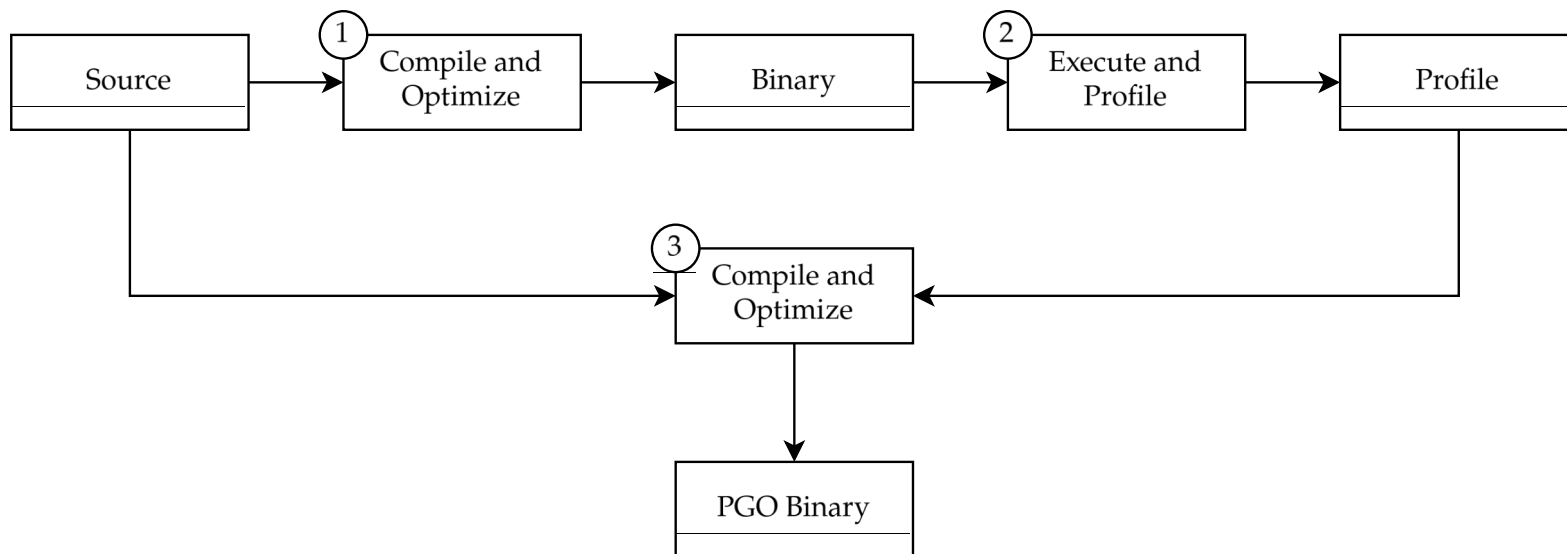
- The optimization process
 - A sequence of transformations (a.k.a. Pipeline)
 - The IR is processed at each pass
 - Output is a *better* program w.r.t. some performance metrics.
 - Supported by static or dynamic analysis that estimate control-flow



Profile Guided Optimization

- A smart optimization technique
 - Leverages *control flow information* captured at runtime to steer optimizations
 - Notable performance improvement
- Control flow information (or *profile*) a.k.a.
 - Weights on control flow edges
 - Counts on basic blocks
- Profiles can be captured via:
 - Instrumentation -> Additional logic inside the program
 - Sampling -> CPU counters + Perf
 - Tracing -> Profile aggregated from the program trace

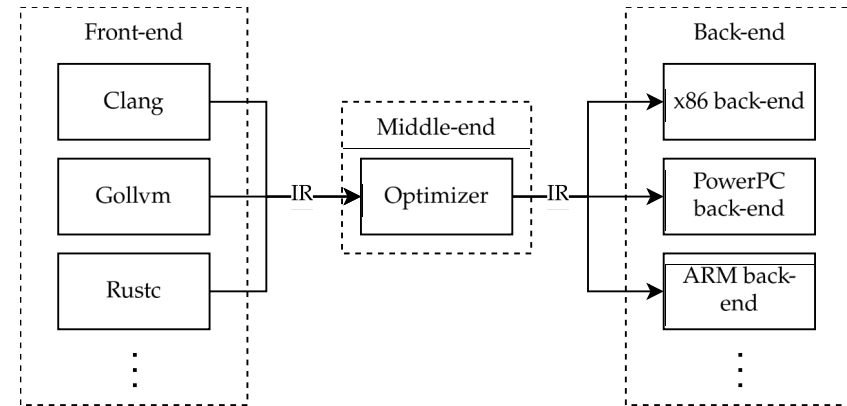
PGO Workflow



The LLVM Project

- The LLVM Project
 - A collection of modular and reusable compiler and toolchain technologies
 - Designed around a modern SSA-based compilation strategy
- Key Components
 - LLVM Core -> Source- and target-independent optimizer
 - Clang -> Native C/C++ compiler for LLVM
 - LLD -> High-performance linker

...and many more!



LLVM architecture

Profile Guided Optimization in LLVM

- Profile information encoded as instruction metadata
 - `branch_weights` -> Counts associated to instructions that change the control flow

```
bb41:
  store i32 0, ptr %i40, align 4
  %i42 = load <4 x i32>, ptr @h, align 16
  br i1 %i39, label %.split.us.preheader, label %.split.preheader, !prof !30
!30 = !{"branch_weights", i32 2000, i32 1000}
```

- `function_entry_counts` -> Number of times a function was called

```
define i32 @foo() !prof !1 {
  ret i32 0
}
!1 = !{"function_entry_count", i64 2590}
```

- `vp` -> Profiles data values passed to functions (Not treated)

Control Flow Analysis

- Profile metadata used to compute additional control flow data
 - BranchProbabilityInfo -> Raw branch weights converted into branch probabilities
 - BlockFrequencyInfo -> Blocks are assigned an execution frequency relative to the entry block
 - ProfileSummaryInfo -> Hotness or coldness of blocks

```
entry:
  %cmp = icmp sgt i32 %x, 0
  br i1 %cmp, label %then, label %else, !prof !0

then:
  ret i32 1

else:
  ret i32 0

!0 = !{"branch_weights", i32 80, i32 20}
!1 = !{"function_entry_count", i64 1000}
```

BranchProbInfo

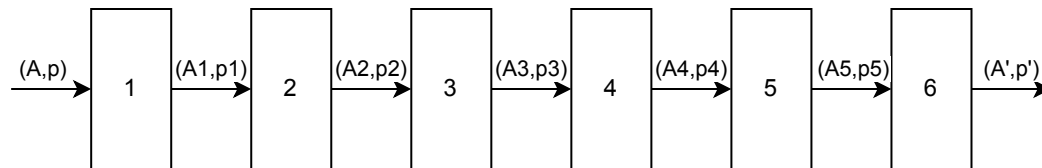
```
edge %entry → %then probability is 0x66666666 / 0x80000000 = 80.00%
edge %entry → %else probability is 0x1999999a / 0x80000000 = 20.00%
```

BlockFrequencyInfo

```
then: float = 0.8, int = 14411518804230144, count = 800
else: float = 0.2, int = 3602879705251840, count = 200
```

Profile Information Propagation

- The starting profile is propagated throughout the entire pipeline
 - Each LLVM optimization is responsible for the update of the profile information
 - Additional logic to handle profile metadata in passes source code



- Bugs in such a logic reduce the efficacy of PGO
 - Subsequent pass will work on wrong profile information
 - A cascading effect will trigger
 - Bad optimization decisions can be taken!

Problem Formalization

Definition 3: Profile propagation analysis problem

Let O be a profile-guided optimization pipeline. Let (G, p) be a profiled program. Let $(G', p') = O(G, p)$ be the profiled program resulting by applying O to (G, p) . The *profile propagation analysis problem* consist of performing the following tasks:

- Determine if O made some profile propagation errors.
- If profile propagation errors were made by O , spot the faulty passes for this errors.

Profile and Block Frequency Formalization

Definition 1: Profiled Program

Given a program A represented as a control-flow graph $G = (E, V)$, a *profile* of A is a function $p : E \rightarrow \mathcal{N}$ that assigns to each edge $(v, w) \in G$ the number of times control flows from basic block v to basic block w . We denote the program A with profile p as the pair (A, p) .

Definition 2: Block Frequency

Give a profiled program $(G = (V, E), p)$ the *block frequency function* is a function $f_p : V \rightarrow \mathcal{N}$ that assigns to each basic block $v \in V$ the number of times v is reached during program execution, as derived from profile p .

Spotting profile propagation errors

Definition 4: Profile Equivalence Relation

Let p and q be two profiles for the same program G and let f_p and f_q be their respective block frequency functions. p is said to be equivalent to q if $\forall v \in V, f_p(v) = f_q(v)$

- Now consider:
 - p -> profile computed by the pipeline on the optimized program
 - q -> profile computed instrumenting and re-executing the optimized program
 - if $p \neq q$ then pipeline made some profile propagation errors

Identifying Culprit Passes

Definition 5: Profile Mismatch

Let p and q be two profiles for the same program G and let f_p and f_q be their respective block frequency functions. Let $p \neq q$.

A *profile mismatch* is a tuple $(G, f, bb, f_p(bb), f_q(bb))$ such that

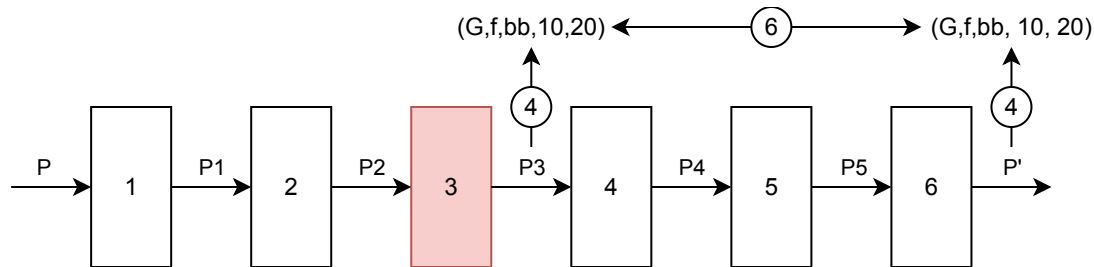
- f is a function within G
 - bb is a basic block within f
 - $f_p(bb) \neq f_q(bb)$
-
- A single pipeline application can results in multiple profile mismatches
 - We need a way to attribute to a mismatch the pass that caused it...

Identifying Culprit Passes

Definition 6: Mismatch Equivalence Relation

Let G and G' be programs, where G' is obtained by manipulating G in some way. Let $m_1 = (G, f, bb, f_p(bb), f_q(bb))$ and $m_2 = (G', f', bb', f_p(bb'), f_q(bb'))$ be two mismatches for their respective programs. Then $m_1 = m_2$ if

- f and f' names are equal
- $f_p(bb) = f_p(bb')$
- $f_q(bb) = f_q(bb')$



Full Methodology